

Name: Solution

MATH 241

SUMMER 2014

QUIZ#3

Problem 1. (5 Points)

Prove that a finite set of nonzero orthogonal vectors form a basis, i.e. they are linearly independent.

Hint: Assume that they are linearly dependent nonzero orthogonal vectors and find a contradiction. Use the fact that the norm of a vector is zero only if the vector is zero.

Problem 2. (5 Points)

Consider the generating set $\{v_1, v_2, v_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$.

Let the vector space $S = \text{span}\{v_1, v_2, v_3\}$

- (a) Is the generating set $\{v_1, v_2, v_3\}$ a linearly independent set of vectors?
- (b) Find a basis for S and the dimension of S .
- (c) Find an orthonormal basis for S using Gram-Schmidt orthogonalization process.
- (d) Find the projection of $u = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ onto vector space S .
- (e) Find the error vector $\epsilon = u - u_p$, where u_p is the projection of u onto S . Verify that ϵ is orthogonal to every vector in S .

① Let $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_n$ a set of nonzero orthogonal set of vectors.
Assume $\underline{v}_j$ is linearly dependent to other vectors, i.e. $\underline{v}_j = \sum_{i \neq j} \alpha_i \underline{v}_i$

Then the norm of $\underline{v}_j$ is $\|\underline{v}_j\|^2 = \underline{v}_j \cdot \underline{v}_j = \underline{v}_j \cdot \sum_{i \neq j} \alpha_i \underline{v}_i = \sum_{i \neq j} \alpha_i \underline{v}_j \cdot \underline{v}_i = \sum_{i \neq j} \alpha_i \cdot 0 = 0$

$\Rightarrow$ But the norm of $\underline{v}_j$ cannot be zero, since only the norm of zero vector is 0. (contradiction) Therefore the nonzero orthogonal vector set cannot be linearly dependent.

② a) No, i.e. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, or use Gauss elimination:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{two pivots, only two vectors in the set is L.I.}$$

b) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$ or $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$ or $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$

c) use the vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$ since they are already orthogonal just normalize them we have $\left\{ \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$

$$d) \underline{u}_p = (\hat{e}_1, \underline{u}) \hat{e}_1 + (\hat{e}_2, \underline{u}) \hat{e}_2$$

$$= \frac{2}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 2/3 \\ -1/3 \end{bmatrix}$$

$$e) \underline{\varepsilon} = \underline{u} - \underline{u}_p = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 5/3 \\ 2/3 \\ -1/3 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 4/3 \\ -2/3 \end{bmatrix} = -\frac{2}{3} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

any vector $\underline{v} \in S$ can be expressed as

$$\underline{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 \quad \text{then} \quad \underline{\varepsilon} \cdot \underline{v} = \alpha_1 \underline{\varepsilon} \cdot \hat{e}_1 + \alpha_2 \underline{\varepsilon} \cdot \hat{e}_2 = \alpha_1 0 + \alpha_2 0 = 0$$

the every vector $\underline{v} \in S$ is orthogonal to error vector $\underline{\varepsilon}$.